Data Preparation and Descriptive Analysis

JFK Airline Delay and Cancellation Operational Risk

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DASE7337 Group Presentation
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Where This Chapter Fits

- The previous chapter covers sector context, core risk factors, SWOT, and PEST.
- This chapter converts BTS raw records into modeling-ready risk assets.
- The key question is not only which charts we produced, but which data definitions can support the next models.

Modeling handoff

The airline-month records are organized into frequency variables, severity variables, and delay-cause variables for distribution fitting, state classification, and aggregate risk simulation.

Data Scope and Cleaning Result

Metric	Value
Raw records	570
Clean records	569
Airport months	72
Airlines	12
Arrival flights	657,671
Delay minutes	11,125,838

- One non-operational record was removed.
- Logical check: delayed arrivals must not exceed total arrivals.
- Airline names were standardized and BTS fields were rewritten into readable analytical names.

Variable Design for Frequency–Severity Modeling

Frequency

- Arrival flights
- 15+ minute delays
- Cancellations and diversions

Severity

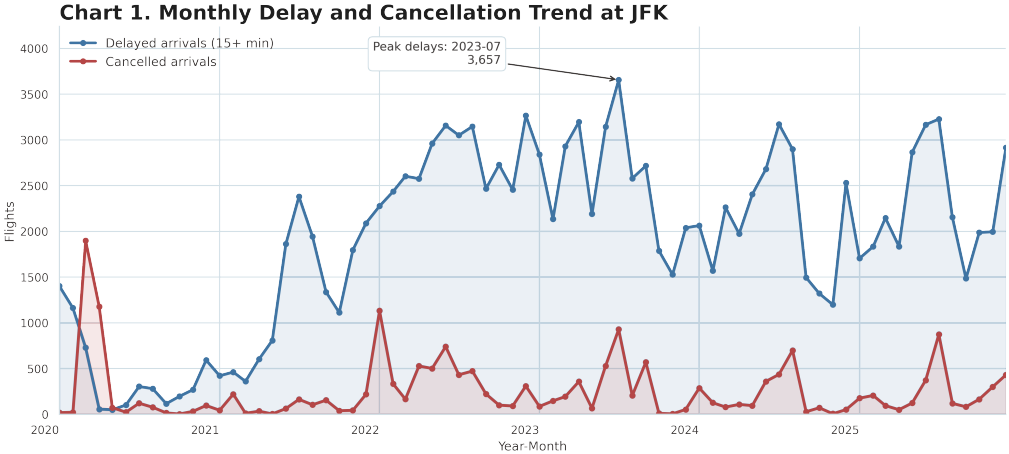
- Total delay minutes
- Average minutes per delay
- Cause-specific minutes

Risk Blocks

- Airline and Late Aircraft
- NAS
- Weather and Security

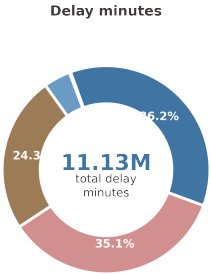
The cleaned structure lets later models ask separately: how often, and how severe?

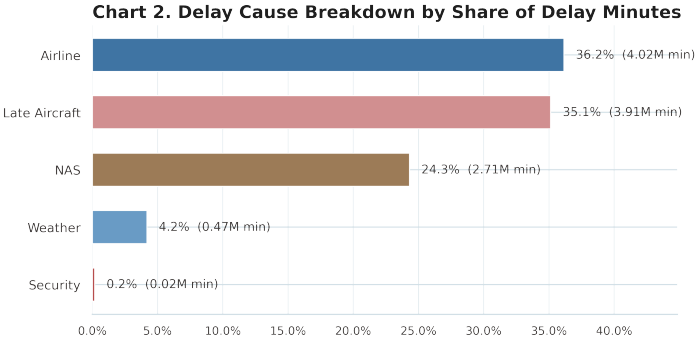
Monthly Trend: Delays and Cancellations Differ



- Peak delay month: July 2023, with 3,657 delayed arrivals and a 32.69% delay rate.
- Peak cancellation month: March 2020, with 1,900 cancellations during the early pandemic shock.

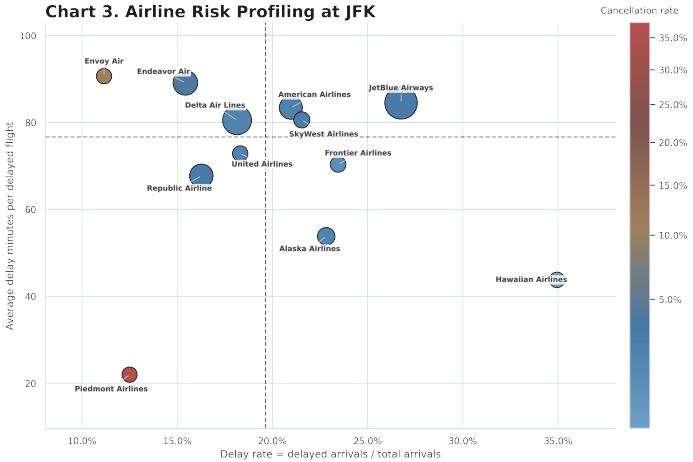
Delay-Cause Structure: Internal Chain Dominates





Airline and Late Aircraft jointly contribute about 71.3% of delay minutes, so daily operational risk is driven mainly by the internal operating chain and system congestion, not only by weather.

Airline Risk Profile and Dashboard Handoff



- JetBlue has the largest total delay minutes.
- Hawaiian has the highest delay rate.
- Envoy Air has the highest average delay severity.
- The offline dashboard supports filtering, interactive charts, and a readable data table.

Open project dashboard

Why This Matters for the Next Models

1. The cleaned core dataset gives clear inputs for frequency and severity models.
2. Cause categories translate BTS fields into operational risk blocks.
3. The monthly time series supports normal/disrupted state classification and Markov transition estimation.
4. The charts and dashboard make the modeling story explainable in business language.